For every fixed concept $i$ and a datapoint $\boldsymbol{z_c}$, maximizing over $j$ is therefore

$$
\begin{aligned}
\arg\max_{j\in[n]} h_{i,j}(\boldsymbol{z_c}) &= \arg\max_{j\in[n]} \left(2\alpha_{i,j}\boldsymbol{d}_i^\intercal(\boldsymbol{z_c}-\boldsymbol{m}_i) - \alpha_{i,j}^2\|\boldsymbol{d}_i\|_2^2\right) \\
&= \arg\max_{j\in[n]} \left(2\alpha_{i,j}\boldsymbol{d}_i^\intercal\left(\sum_{\ell=1}^{k}\alpha_{\ell,c_\ell}\boldsymbol{d}_\ell - \boldsymbol{m}_i\right) - \alpha_{i,j}^2\|\boldsymbol{d}_i\|_2^2\right) \\
&= \arg\max_{j\in[n]} \left(2\alpha_{i,j}\boldsymbol{d}_i^\intercal\left(\alpha_{i,c_i}\boldsymbol{d}_i + \sum_{\ell\neq i}\alpha_{\ell,c_\ell}\boldsymbol{d}_\ell - \boldsymbol{m}_i\right) - \alpha_{i,j}^2\|\boldsymbol{d}_i\|_2^2\right) \\
&= \arg\max_{j\in[n]} \left(2\alpha_{i,j}\alpha_{i,c_i}\|\boldsymbol{d}_i\|_2^2 - \alpha_{i,j}^2\|\boldsymbol{d}_i\|_2^2\right) \\
&= \arg\max_{j\in[n]} \left(-(\alpha_{i,j}-\alpha_{i,c_i})^2\|\boldsymbol{d}_i\|_2^2\right) \\
&= \arg\min_{j\in[n]} (\alpha_{i,j}-\alpha_{i,c_i})^2 \\
&= c_i.
\end{aligned}
$$

where we dropped the additive term $2\boldsymbol{m}_i^\intercal\boldsymbol{z_c} - \|\boldsymbol{m}_i\|_2^2$, since it does not depend on $j$, and used orthogonality: $\boldsymbol{d}_i^\intercal\boldsymbol{d}_\ell = 0$ for $\ell \neq i$ and $\boldsymbol{d}_i^\intercal\boldsymbol{m}_i = 0$. The last step follows since $\|\boldsymbol{d}_i\|_2^2 > 0$ is constant and $\{\alpha_{i,j}\}_{j=1}^n$ are distinct.

As shown above, such a classifier is easy to construct and correctly classifies all the points in the concept space. If orthogonality did not hold, the nearest neighbor classifier is not guaranteed to be correct. We illustrate the geometry of the probes together with the decision regions in Fig. 16 in two cases, where $k = 2, n = 30$ and the deviation from orthogonal features varies slightly. We note that the weight vectors, indicated as arrows in the plot, are near-parallel.

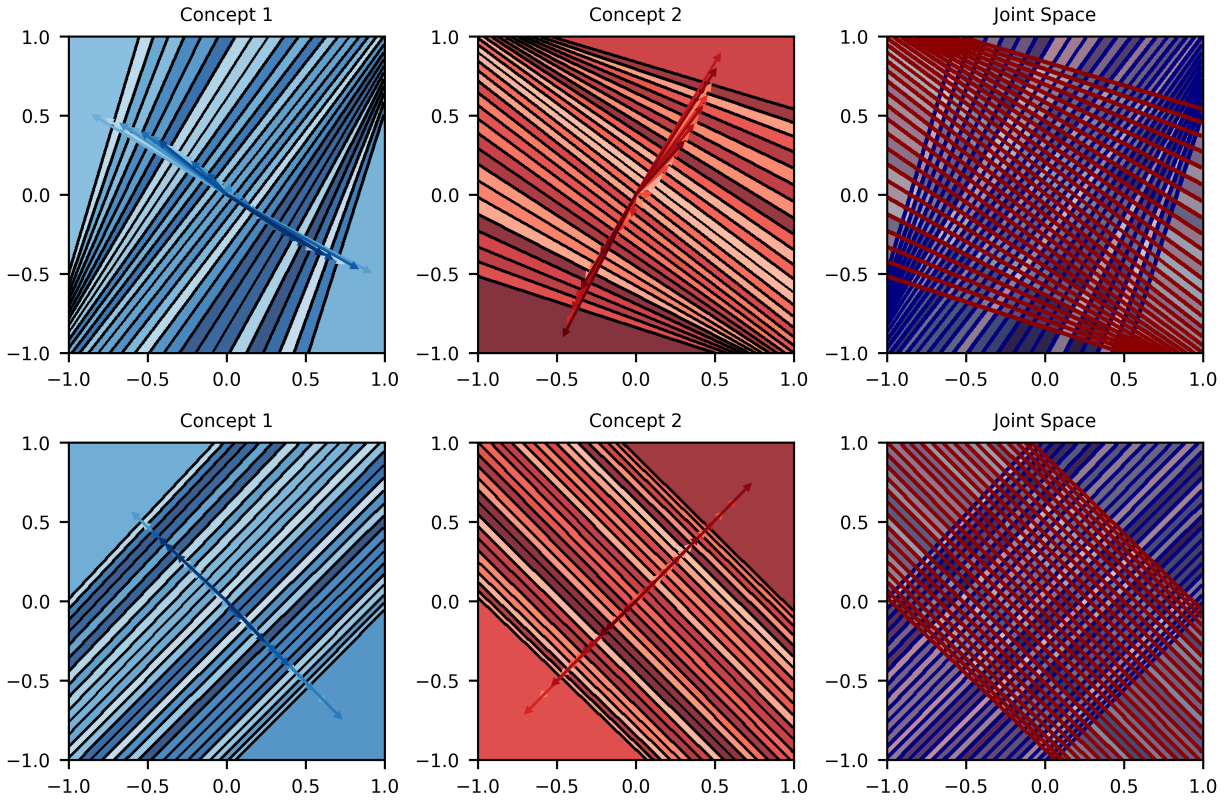


*Figure 16.* **Probe geometry in the LRH toy setup ($k = 2$, $n = 30$).** Each row corresponds to a different choice of concept directions (top: mildly non-orthogonal; bottom: closer to orthogonal). Left and middle columns show concept-wise decision regions for concept 1 (blue) and concept 2 (red), with arrows indicating probe directions. The right column overlays both concept families in the joint space; intersections of blue and red stripes form the cells associated with tuples $(c_1, c_2)$.

### F.2. Case 2: Ideal "on-off" concept classifier

Here we consider a setting that intuitively could be described as exhibiting behavior that is desirable for compositional generalization. In particular, we consider a CLIP-style linearly compositional model with the constraint that the probe scores $h_{i,j}$ should be constant among the matches, and constant among the mis-matches.

In CLIP-like models, this is usually viewed through cosine similarity. In the normalized view ($\|\boldsymbol{z}\|_2 = 1$ and $\|\boldsymbol{w}_{i,j}\|_2 = 1$), decision regions on the sphere are spherical caps rather than Euclidean half-spaces. For a cosine-similarity classifier, the decision region for class $(i, j)$ is

$$\mathcal{R}_{i,j} := \left\{ \boldsymbol{z} \in \mathbb{S}^{d-1} : \boldsymbol{w}_{i,j}^\intercal \boldsymbol{z} > \boldsymbol{w}_{i,k}^\intercal \boldsymbol{z} \; \forall k \neq j \right\}. \tag{72}$$

We refer to this setting as "on-off" concept classifier, as there are only two possible scores for each probe: $\alpha$ if the concept matches, and $\beta$ if it does not. That is, exist constants $\alpha > \beta$ in $[-1, 1]$ such that for all $i, j$ and all tuples $\boldsymbol{c}$,

$$\boldsymbol{w}_{i,j}^\intercal \boldsymbol{z}_{\boldsymbol{c}} = \begin{cases} \alpha & \text{if } j = c_i \\ \beta & \text{if } j \neq c_i. \end{cases} \tag{73}$$

We illustrate this setting in Fig. 17.

The key results in this section are two-fold: (1) a model exhibiting this behavior requires at least $1 + k(n-1)$ dimensions, meaning that under a large number of concepts and values per concept, even current systems will not be able to reliably distinguish between correct and incorrect matches between a sample and its corresponding concept values (Proposition 6); (2) assuming that the model is able to distinguish between correct and incorrect matches, one can linearly approximate the representations in a way that preserves the "on-off" pattern (Proposition 7).

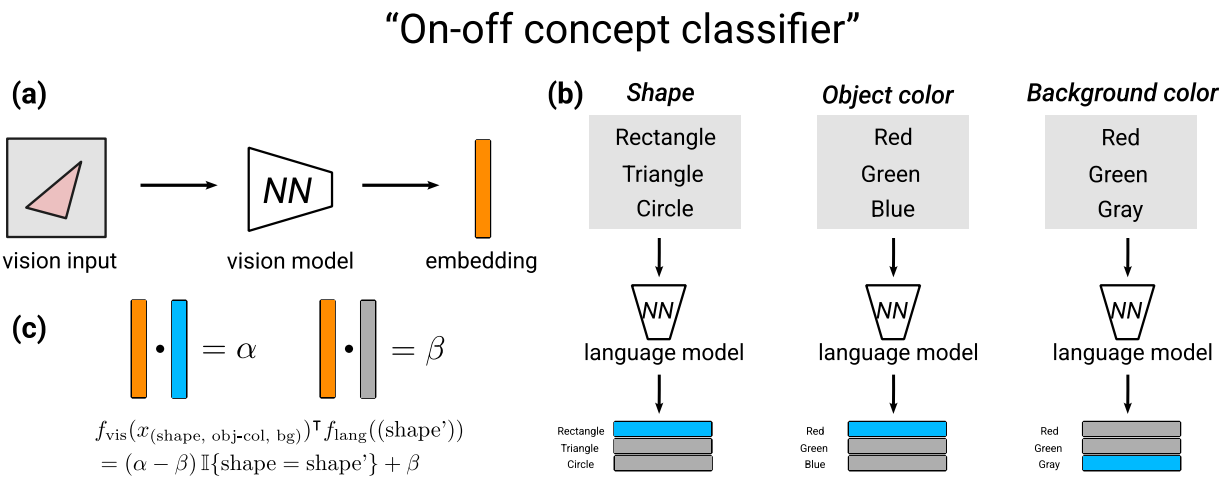


*Figure 17.* **Illustration of the "on-off concept classifier" mechanism. (a)** A vision input is processed by a neural network to produce an embedding. **(b)** Each concept (e.g., shape, object color, background color) is probed independently using a set of language model probes, one per possible value. **(k)** The probe for a given concept yields a high score $\alpha$ if the concept matches and a lower score $\beta$ otherwise, as formalized in the logit equation at bottom.

**Proposition 6** (Minimal dimensionality from fixed dot-products)**.** Assume $k \geq 2$ and $n \geq 2$. For each concept $i \in [k]$ and value $j \in [n]$, let the probe weight be$\boldsymbol{w}_{i,j} \in \mathbb{R}^d$, such that for any tuple $\boldsymbol{c} = (c_1, \ldots, c_k) \in [n]^k$ there exists a corresponding $\boldsymbol{z}_{\boldsymbol{c}} \in \mathbb{R}^d$, and there exist $\alpha, \beta \in \mathbb{R}$ with $\alpha \neq \beta$ such that the "on-off" pattern holds (and $\alpha \neq -\beta(n-1)$):

$$\boldsymbol{w}_{i,j}^\intercal \boldsymbol{z}_{\boldsymbol{c}} = \begin{cases} \alpha, & j = c_i, \\ \beta, & j \neq c_i, \end{cases} \qquad \text{for all } i, j, \boldsymbol{c}, \tag{74}$$

Then the ambient dimension $d$ must satisfy

$$d \geq 1 + k(n-1). \tag{75}$$

Moreover, this bound is tight: for any $(\alpha, \beta)$ with $\alpha \neq \beta$, there exist explicit probe/representation families that realize (74) in dimension $d = 1 + k(n-1)$.

*Proof.* We write the constraints in matrix form. First, we stack probe vectors as rows:

$$\boldsymbol{P} = \begin{bmatrix} \boldsymbol{w}_{1,1}^\intercal \\ \vdots \\ \boldsymbol{w}_{k,n}^\intercal \end{bmatrix} \in \mathbb{R}^{kn \times d}, \qquad (\text{row } (i-1)n + j = \boldsymbol{w}_{i,j}^\intercal). \tag{76}$$

Then stack representations as columns:

$$\boldsymbol{X} = \begin{bmatrix} \boldsymbol{z}_{\boldsymbol{c}_1} & \cdots & \boldsymbol{z}_{\boldsymbol{c}_{n^k}} \end{bmatrix} \in \mathbb{R}^{d \times n^k}. \tag{77}$$

The "on-off" constraints (74) become

$$\boldsymbol{Y} = \boldsymbol{P}\boldsymbol{X} \in \mathbb{R}^{kn \times n^k}, \tag{78}$$

where entries of $\boldsymbol{Y}$ are

$$y_{(i,j),\boldsymbol{c}} = \begin{cases} \alpha, & j = c_i, \\ \beta, & j \neq c_i. \end{cases}, \tag{79}$$

and will sometimes denote rows of $\boldsymbol{Y}$ as $\boldsymbol{y}_{(i,j)}$ instead. To understand the rank of $\boldsymbol{Y}$, first note that it is at most $kn$. Additionally, note that the rows can be divided into $k$ blocks. As we will show, each such block contains up to one dependence that is achieved by summing up the rows within a block.

For each block $i \in [k]$ and any column $\boldsymbol{c} \in [n]^k$, we can sum the $n$ entries in that block as

$$\sum_{j=1}^{n} y_{(i,j),\boldsymbol{c}} = \sum_{j=1}^{n} \big(\alpha \mathbb{I}_{c_i = j} + \beta \mathbb{I}_{c_i \neq j}\big) = \alpha + \beta(n-1). \tag{80}$$

Importantly, this sum is the same regardless of $\boldsymbol{c}$ and chosen $i$. Therefore, each block sums up to the same vector, and is therefore redundant. This implies that the rank can be at most $nk - (k-1) = 1 + k(n-1)$. Next, we show that within each block, each row is independent.

For that, we will show that for any concept "block" $i \in [k]$ every row $(i,j)$ corresponding to the value $j$ of the $i$-th concept is linearly independent of all the rows in blocks $i' \neq i$. We denote the span of the other blocks' rows as

$$\mathcal{S} := \text{span}(\{\boldsymbol{y}_{(i',j')}\}_{i' \neq i, j' \in [n]}). \tag{81}$$

For that, we take any two columns $\boldsymbol{c}, \boldsymbol{c}' \in [n]^k$ that are identical up to the $i$-th concept and satisfy $c_m = c'_m$ for $m \neq i$ and $c_i \neq c'_i$, and note that all the rows outside the $i$-th block are identical: $y_{(i',j'),\boldsymbol{c}} = y_{(i',j'),\boldsymbol{c}'}$ for all $i' \neq i$ and all $j' \in [n]$, and only differ in the $i$-th block rows. Because of that, any linear combination from $\mathcal{S}$ with some coefficients will have matching columns:

$$\Big(\sum \lambda_n \boldsymbol{s}_n\Big)_{\boldsymbol{c}} = \Big(\sum \lambda_n \boldsymbol{s}_n\Big)_{\boldsymbol{c}'}, \tag{82}$$

but both $\boldsymbol{c}$ and $\boldsymbol{c}'$ differ in exactly the $i$-th concept, and therefore their column values in the $i$-th block must be different:

$$y_{(i,c_i),\boldsymbol{c}} \neq y_{(i,c_i),\boldsymbol{c}'}, \tag{83}$$

and therefore no vector in the span can be equal to $\boldsymbol{y}_{i,c_i}$. All that is needed to show now is that taking any linear combination of the rows in the $i$-th block cannot be expressed by any $\boldsymbol{c} \in \mathcal{S}$.

We do this by applying the same reasoning but instead of taking 2 rows we take $n$ rows. Concretely, for the $i$-th concept we take all $n$ rows: $(i,1), \ldots, (i,n)$ and consider $\boldsymbol{c}_1, \ldots, \boldsymbol{c}_n$ which are all identical up to the $i$-th concept. Again, for any $i' \neq i$ and any row index $j \in [n]$, the values are constant across these columns: $y_{(i',j),\boldsymbol{c}_m} = y_{(i',j),\boldsymbol{c}_{m'}}$ for all $m, m' \in [n]$, and only the $i$-th block rows change across $\boldsymbol{c}_1, \ldots, \boldsymbol{c}_n$. We can thus consider the span of the $i$-th block among the $n$ rows by taking $\boldsymbol{\lambda} \in \mathbb{R}^n$ and considering only the $n$ columns corresponding to $\boldsymbol{c}_1, \ldots, \boldsymbol{c}_n$, writing the linear combination as

$$\boldsymbol{r} := \sum_{j=1}^{n} \lambda_j \boldsymbol{y}_{(i,j)} \in \mathbb{R}^{n^k}. \tag{84}$$

We now evaluate $\boldsymbol{r}$ on the $n$ columns $\boldsymbol{c}_1, \ldots, \boldsymbol{c}_n$. By construction, in column $\boldsymbol{c}_m$ the only row in block $i$ that takes value $\alpha$ is $(i,m)$, while all other $(i,j)$ take value $\beta$. Therefore the $m$-th selected entry is

$$r_m = \lambda_m \alpha + \sum_{j \neq m} \lambda_j \beta = \lambda_m \alpha + (S - \lambda_m)\beta, \tag{85}$$